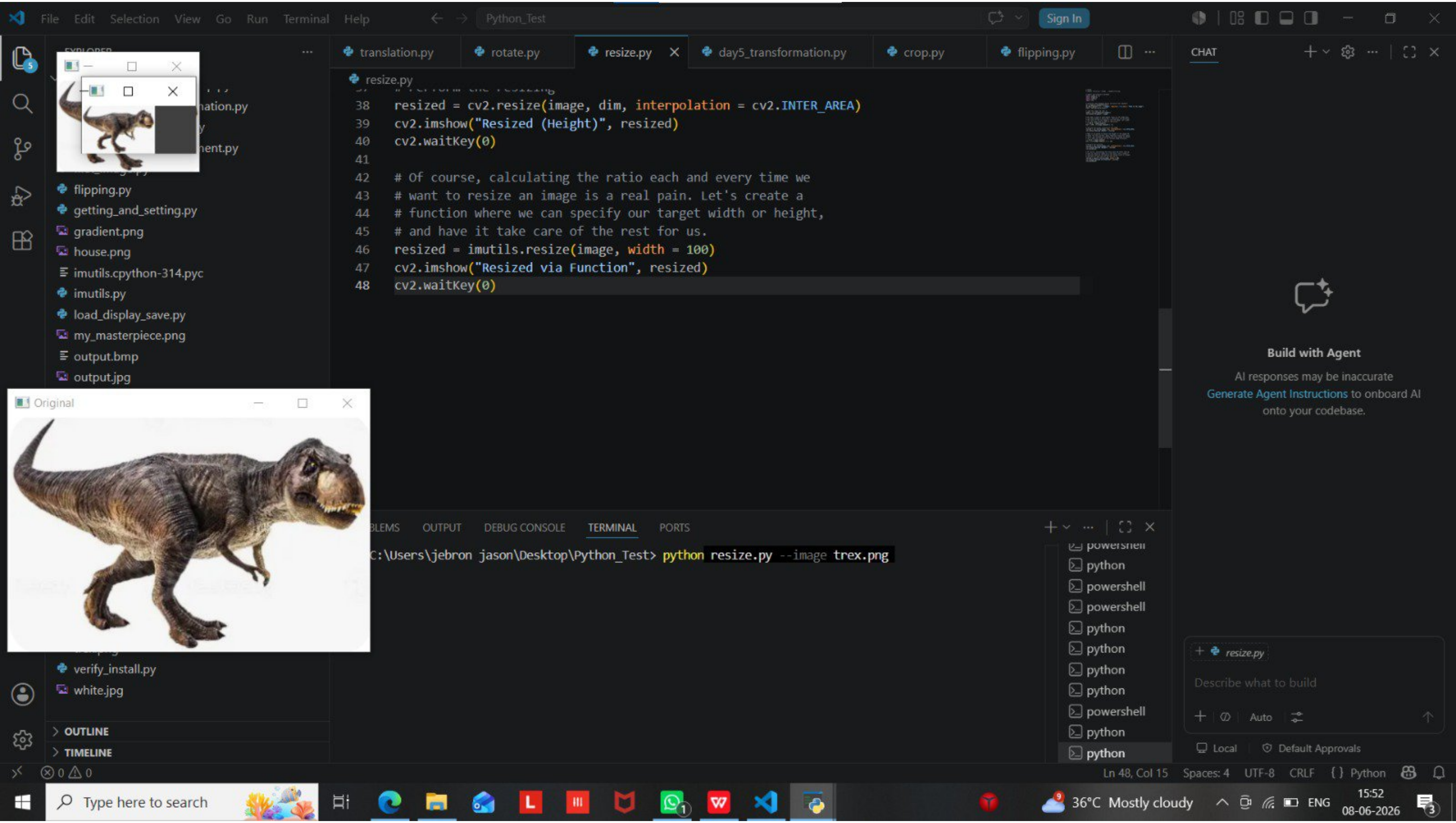
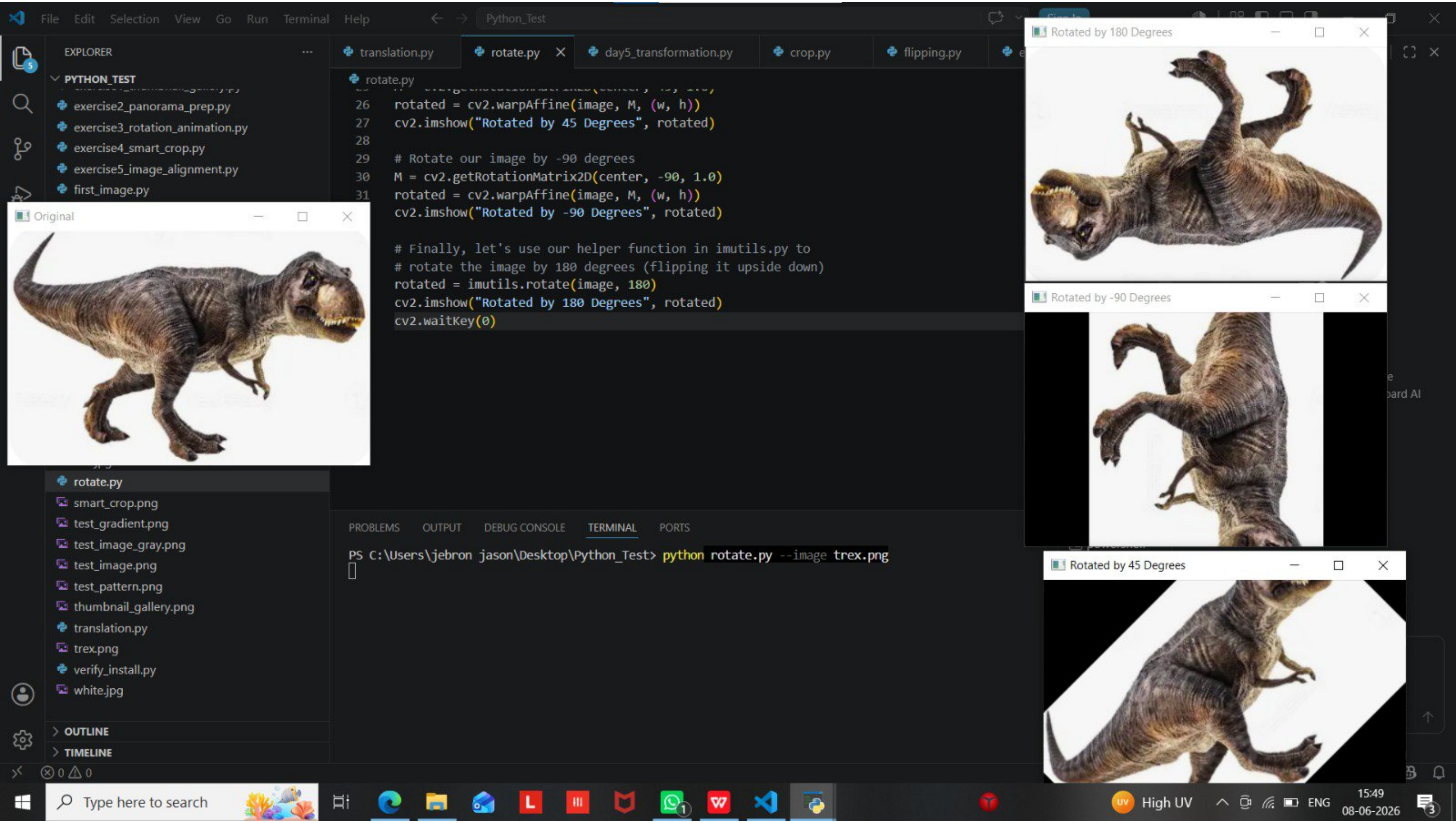
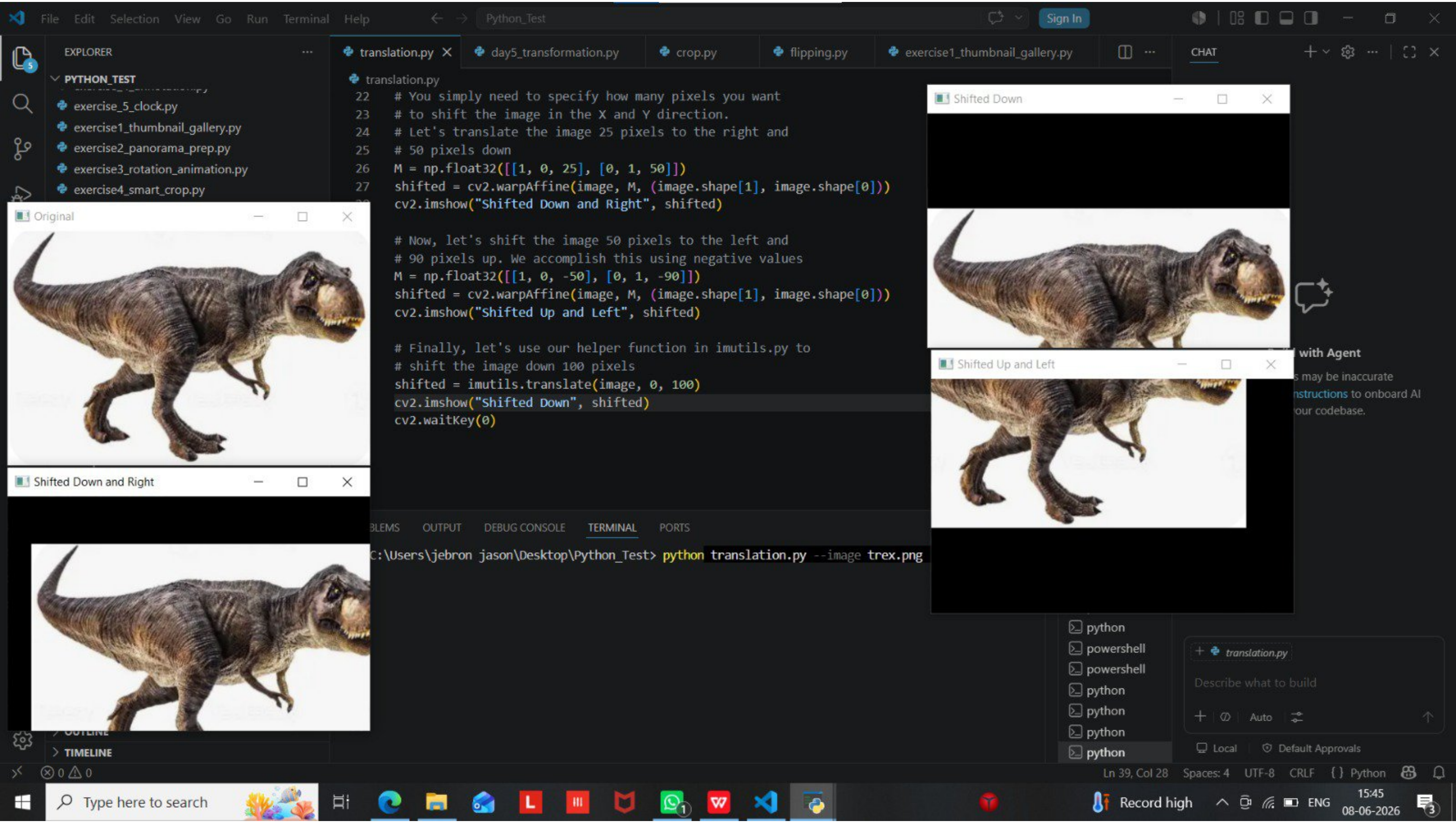
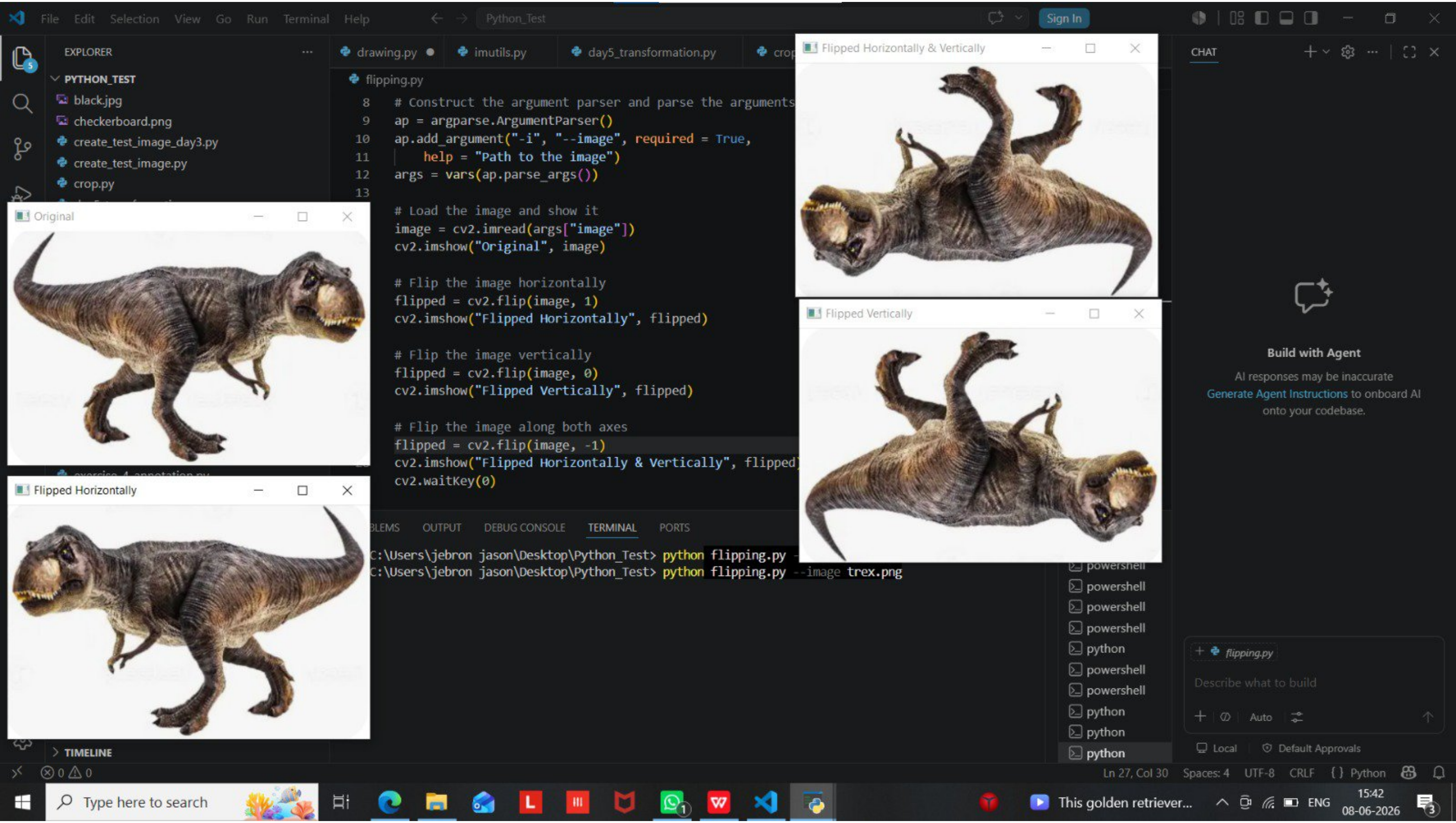
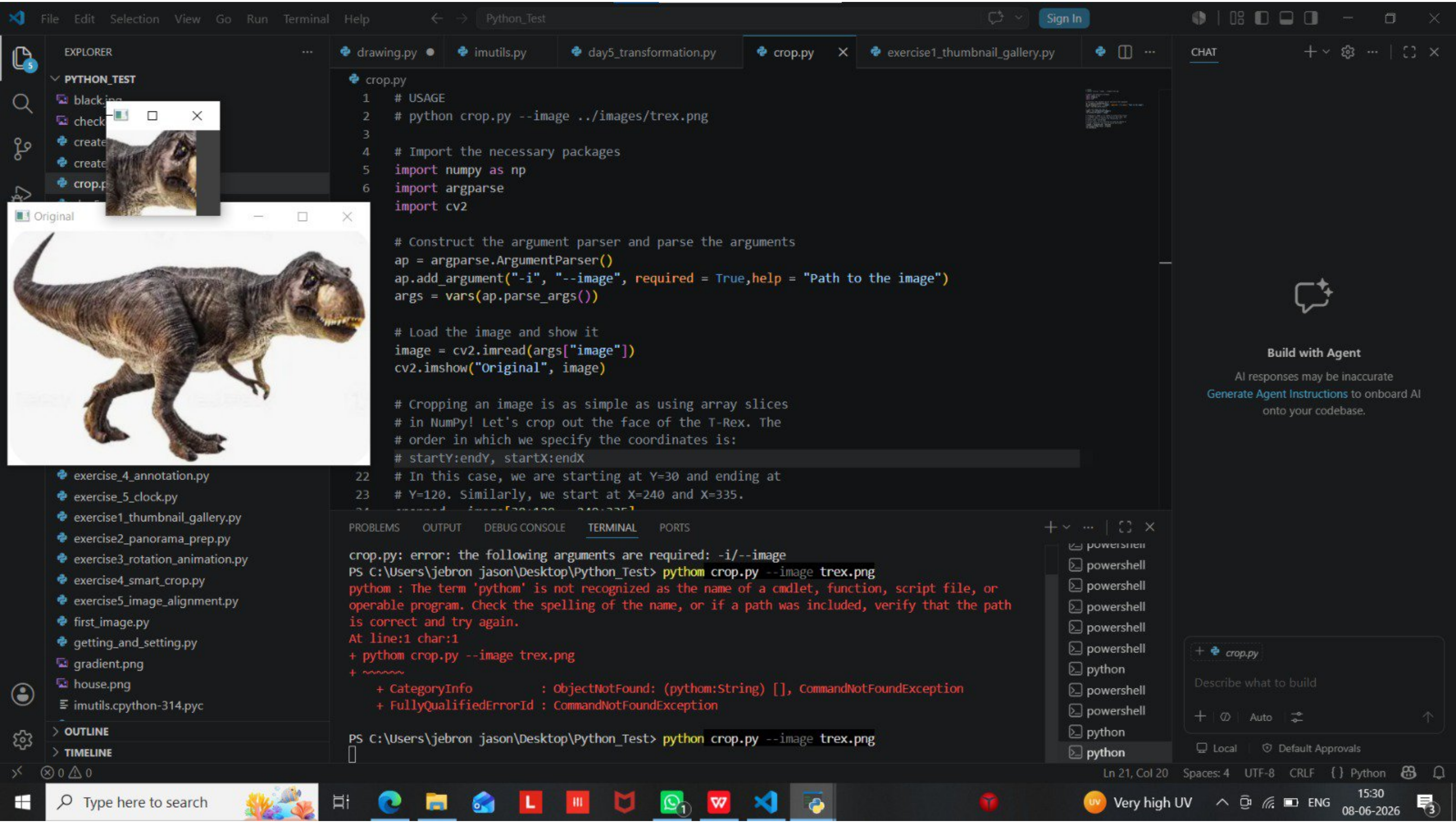

File Edit Selection View Go Run Terminal Help
Python_Test
Sign In
translation.py
rotate.py
resize.py
day5_transformation.py
crop.py
flipping.py
CHAT
resize.py
38 resized = cv2.resize(image, dim, interpolation = cv2.INTER_AREA)
39 cv2.imshow("Resized (Height)", resized)
40 cv2.waitKey(0)
41
42 # Of course, calculating the ratio each and every time we
43 # want to resize an image is a real pain. Let's create a
44 # function where we can specify our target width or height,
45 # and have it take care of the rest for us.
46 resized = imutils.resize(image, width = 100)
47 cv2.imshow("Resized via Function", resized)
48 cv2.waitKey(0)
flipping.py
getting_and_setting.py
gradient.png
house.png
imutils.cpython-314.pyc
imutils.py
load_display_save.py
my_masterpiece.png
output.bmp
output.jpg
Original
verify_install.py
white.jpg
OUTLINE
TIMELINE
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
C:\Users\jebron jason\Desktop\Python_Test> python resize.py --image trex.png
python
powershell
powershell
python
python
python
python
powershell
python
python
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
Describe what to build
Auto
Local
Default Approvals
Ln 48, Col 15 Spaces: 4 UTF-8 CRLF { } Python
Type here to search
36°C Mostly cloudy
ENG
15:52
08-06-2026


File Edit Selection View Go Run Terminal Help
Python_Test
EXPLORER
PYTHON_TEST
exercise2_panorama_prep.py
exercise3_rotation_animation.py
exercise4_smart_crop.py
exercise5_image_alignment.py
first_image.py
rotate.py
smart_crop.png
test_gradient.png
test_image_gray.png
test_image.png
test_pattern.png
thumbnail_gallery.png
translation.py
trex.png
verify_install.py
white.jpg
OUTLINE
TIMELINE
translation.py
rotate.py
day5_transformation.py
crop.py
flipping.py
rotated = cv2.warpAffine(image, M, (w, h))
cv2.imshow("Rotated by 45 Degrees", rotated)
# Rotate our image by -90 degrees
M = cv2.getRotationMatrix2D(center, -90, 1.0)
rotated = cv2.warpAffine(image, M, (w, h))
cv2.imshow("Rotated by -90 Degrees", rotated)
# Finally, let's use our helper function in imutils.py to
# rotate the image by 180 degrees (flipping it upside down)
rotated = imutils.rotate(image, 180)
cv2.imshow("Rotated by 180 Degrees", rotated)
cv2.waitKey(0)
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\jebron jason\Desktop\Python_Test> python rotate.py --image trex.png
Original
Rotated by 180 Degrees
Rotated by -90 Degrees
Rotated by 45 Degrees
Type here to search
High UV
ENG
15:49
08-06-2026

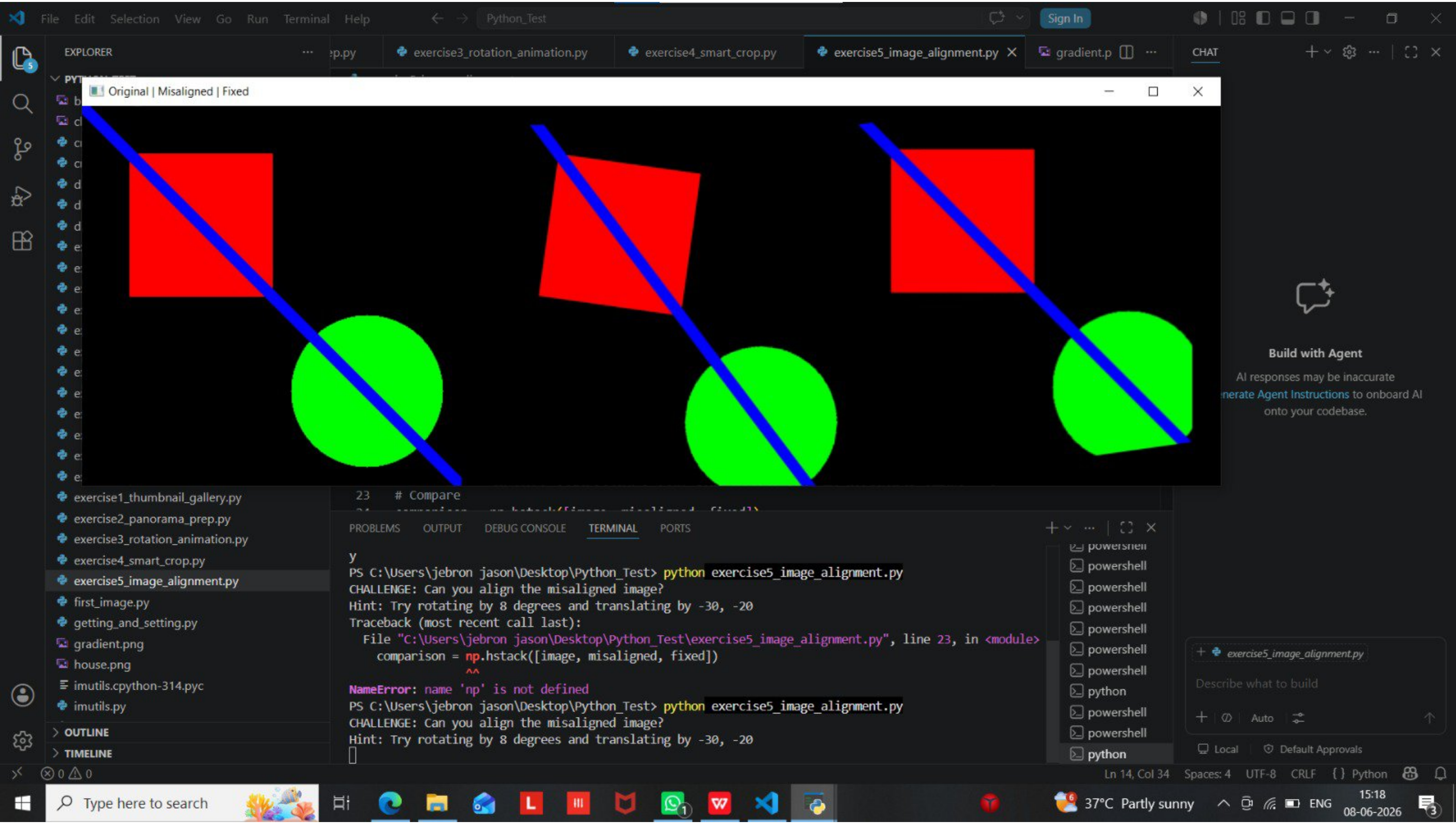

File Edit Selection View Go Run Terminal Help
Python_Test
Sign In
EXPLORER
PYTHON_TEST
exercise_5_clock.py
exercise1_thumbnail_gallery.py
exercise2_panorama_prep.py
exercise3_rotation_animation.py
exercise4_smart_crop.py
translation.py
day5_transformation.py
crop.py
flipping.py
exercise1_thumbnail_gallery.py
CHAT
# You simply need to specify how many pixels you want
# to shift the image in the X and Y direction.
# Let's translate the image 25 pixels to the right and
# 50 pixels down
M = np.float32([[1, 0, 25], [0, 1, 50]])
shifted = cv2.warpAffine(image, M, (image.shape[1], image.shape[0]))
cv2.imshow("Shifted Down and Right", shifted)
# Now, let's shift the image 50 pixels to the left and
# 90 pixels up. We accomplish this using negative values
M = np.float32([[1, 0, -50], [0, 1, -90]])
shifted = cv2.warpAffine(image, M, (image.shape[1], image.shape[0]))
cv2.imshow("Shifted Up and Left", shifted)
# Finally, let's use our helper function in imutils.py to
# shift the image down 100 pixels
shifted = imutils.translate(image, 0, 100)
cv2.imshow("Shifted Down", shifted)
cv2.waitKey(0)
Original
Shifted Down and Right
Shifted Down
Shifted Up and Left
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
C:\Users\jebron jason\Desktop\Python_Test> python translation.py --image trex.png
python
powershell
powershell
python
python
python
python
with Agent
Describe what to build
Auto
Local
Default Approvals
TIMELINE
Ln 39, Col 28
Spaces: 4
UTF-8
CRLF
Python
Type here to search
Record high
ENG
15:45
08-06-2026

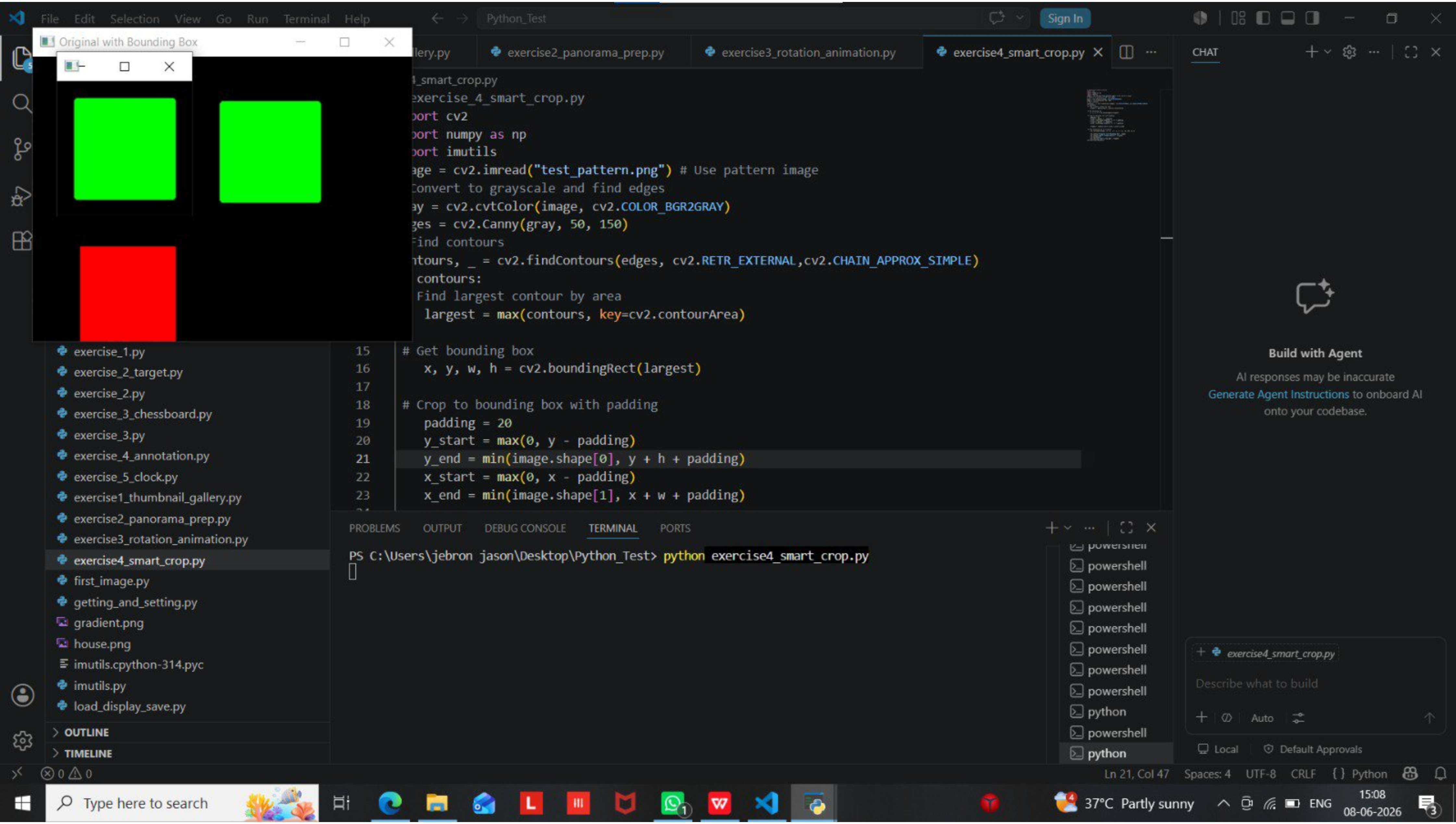


File Edit Selection View Go Run Terminal Help
Python_Test
Sign In
EXPLORER
PYTHON_TEST
black.jpg
checkerboard.png
create_test_image_day3.py
create_test_image.py
crop.py
drawing.py
imutils.py
day5_transformation.py
flipping.py
# Construct the argument parser and parse the arguments
ap = argparse.ArgumentParser()
ap.add_argument("-i", "--image", required = True,
help = "Path to the image")
args = vars(ap.parse_args())
# Load the image and show it
image = cv2.imread(args["image"])
cv2.imshow("Original", image)
# Flip the image horizontally
flipped = cv2.flip(image, 1)
cv2.imshow("Flipped Horizontally", flipped)
# Flip the image vertically
flipped = cv2.flip(image, 0)
cv2.imshow("Flipped Vertically", flipped)
# Flip the image along both axes
flipped = cv2.flip(image, -1)
cv2.imshow("Flipped Horizontally & Vertically", flipped)
cv2.waitKey(0)
Original
Flipped Horizontally
Flipped Horizontally & Vertically
Flipped Vertically
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
C:\Users\jebron jason\Desktop\Python_Test> python flipping.py
C:\Users\jebron jason\Desktop\Python_Test> python flipping.py --image trex.png
CHAT
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
Describe what to build
Auto
Local
Default Approvals
TIMELINE
Ln 27, Col 30
Spaces: 4
UTF-8
CRLF
Python
Type here to search
This golden retriever...
ENG
15:42
08-06-2026

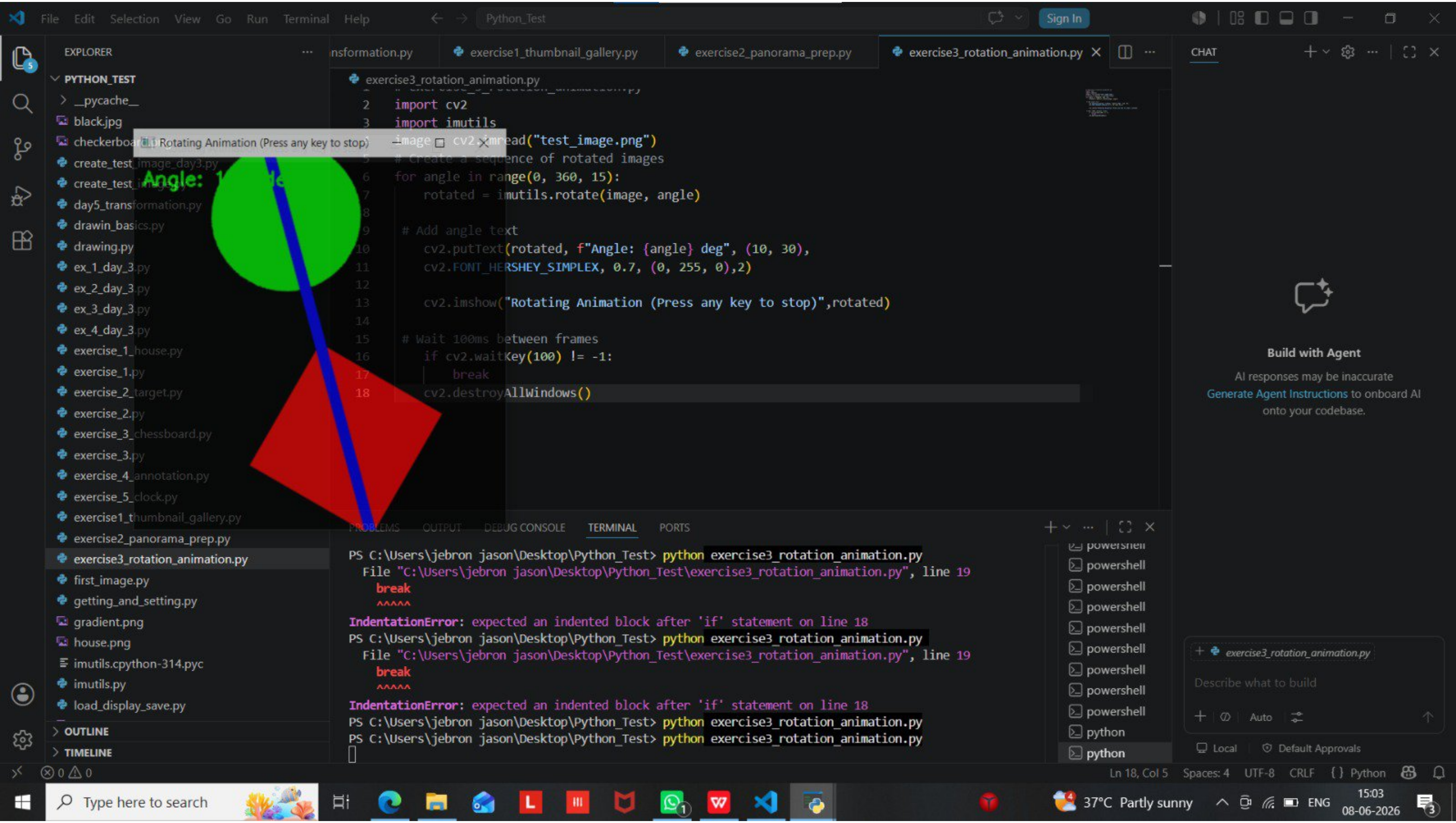


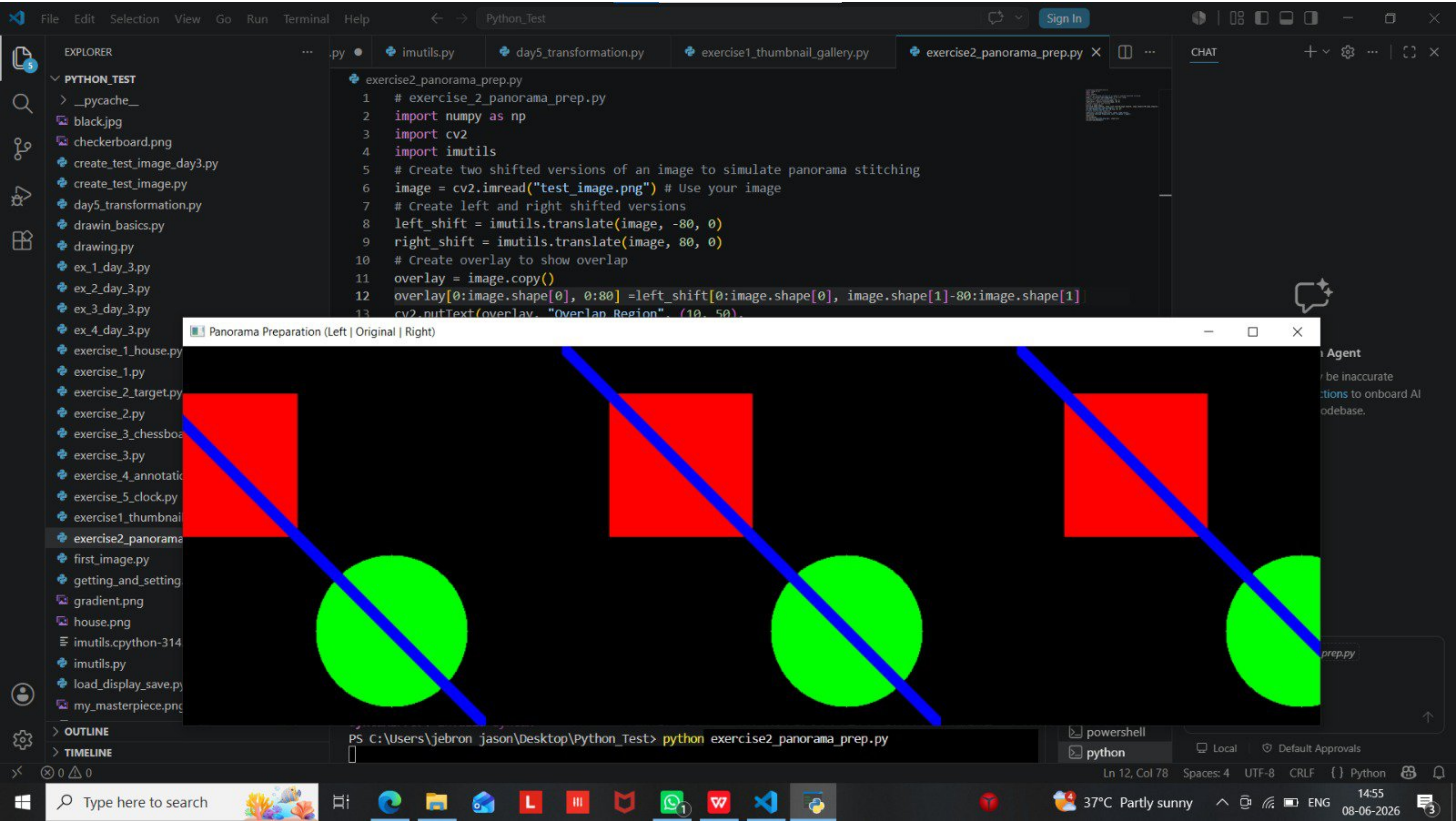
crop.py
# USAGE
# python crop.py --image ../images/trex.png
# Import the necessary packages
import numpy as np
import argparse
import cv2
# Construct the argument parser and parse the arguments
ap = argparse.ArgumentParser()
ap.add_argument("-i", "--image", required = True,help = "Path to the image")
args = vars(ap.parse_args())
# Load the image and show it
image = cv2.imread(args["image"])
cv2.imshow("Original", image)
# Cropping an image is as simple as using array slices
# in NumPy! Let's crop out the face of the T-Rex. The
# order in which we specify the coordinates is:
# startY:endY, startX:endX
# In this case, we are starting at Y=30 and ending at
# Y=120. Similarly, we start at X=240 and X=335.
Original
crop.py: error: the following arguments are required: -i/--image
PS C:\Users\jebron jason\Desktop\Python_Test> python crop.py --image trex.png
python : The term 'python' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a path was included, verify that the path is correct and try again.
At line:1 char:1
+ python crop.py --image trex.png
+ ~~~~~~
+ CategoryInfo : ObjectNotFound: (python:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException
PS C:\Users\jebron jason\Desktop\Python_Test> python crop.py --image trex.png
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.


Original | Misaligned | Fixed
EXPLORER
exercise3_rotation_animation.py
exercise4_smart_crop.py
exercise5_image_alignment.py
gradient.p
CHAT
Build with Agent
AI responses may be inaccurate
onto your codebase.
exercise1_thumbnail_gallery.py
exercise2_panorama_prep.py
exercise3_rotation_animation.py
exercise4_smart_crop.py
exercise5_image_alignment.py
first_image.py
getting_and_setting.py
gradient.png
house.png
imutils.cpython-314.pyc
imutils.py
OUTLINE
TIMELINE
23 # Compare
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
y
PS C:\Users\jebron jason\Desktop\Python_Test> python exercise5_image_alignment.py
CHALLENGE: Can you align the misaligned image?
Hint: Try rotating by 8 degrees and translating by -30, -20
Traceback (most recent call last):
File "C:\Users\jebron jason\Desktop\Python_Test\exercise5_image_alignment.py", line 23, in <module>
comparison = np.hstack([image, misaligned, fixed])
^^
NameError: name 'np' is not defined
PS C:\Users\jebron jason\Desktop\Python_Test> python exercise5_image_alignment.py
CHALLENGE: Can you align the misaligned image?
Hint: Try rotating by 8 degrees and translating by -30, -20
powershell
python
Describe what to build
Auto
Local
Default Approvals
Ln 14, Col 34 Spaces: 4 UTF-8 CRLF { } Python
Type here to search
37°C Partly sunny
ENG
15:18
08-06-2026

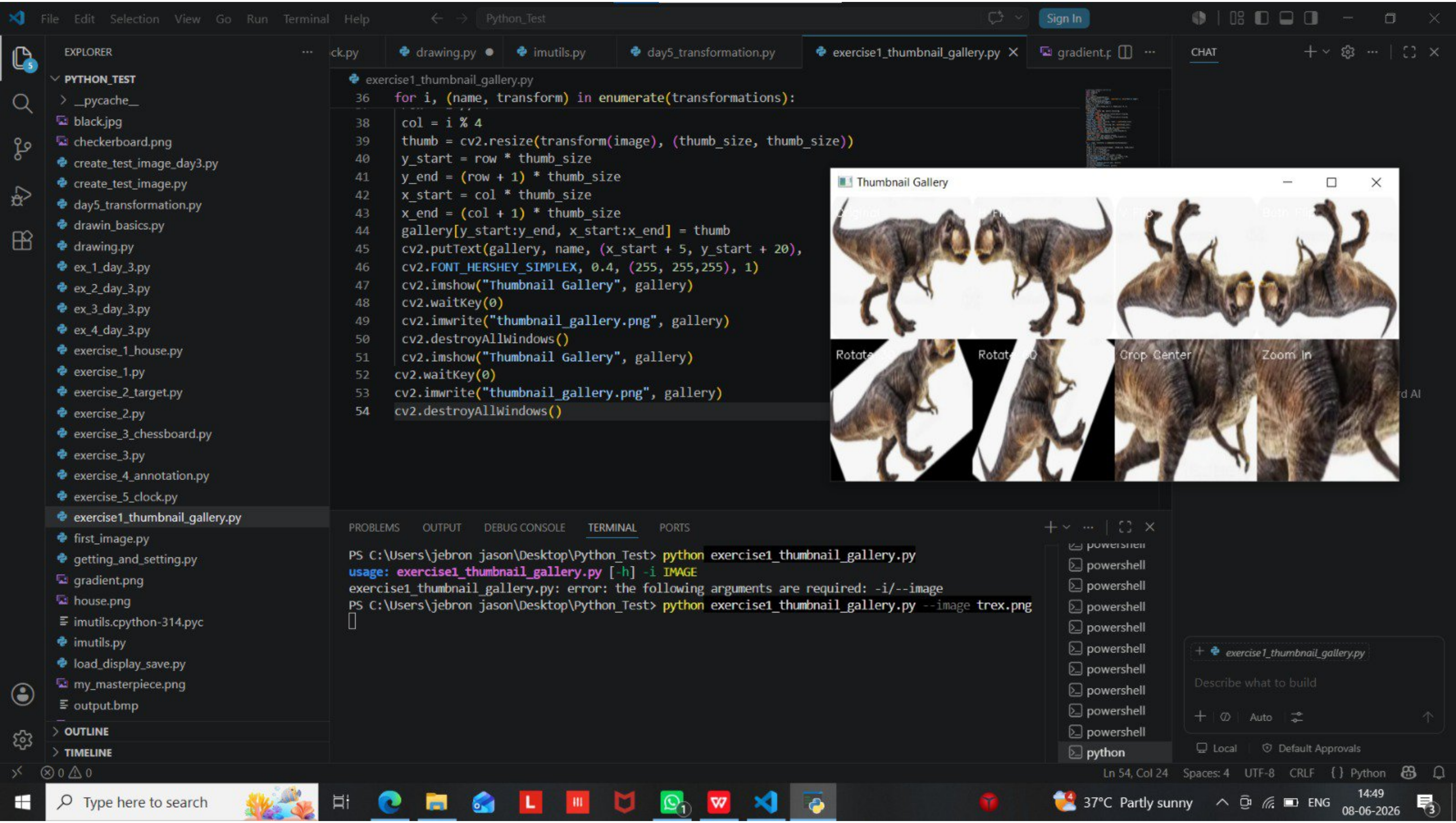


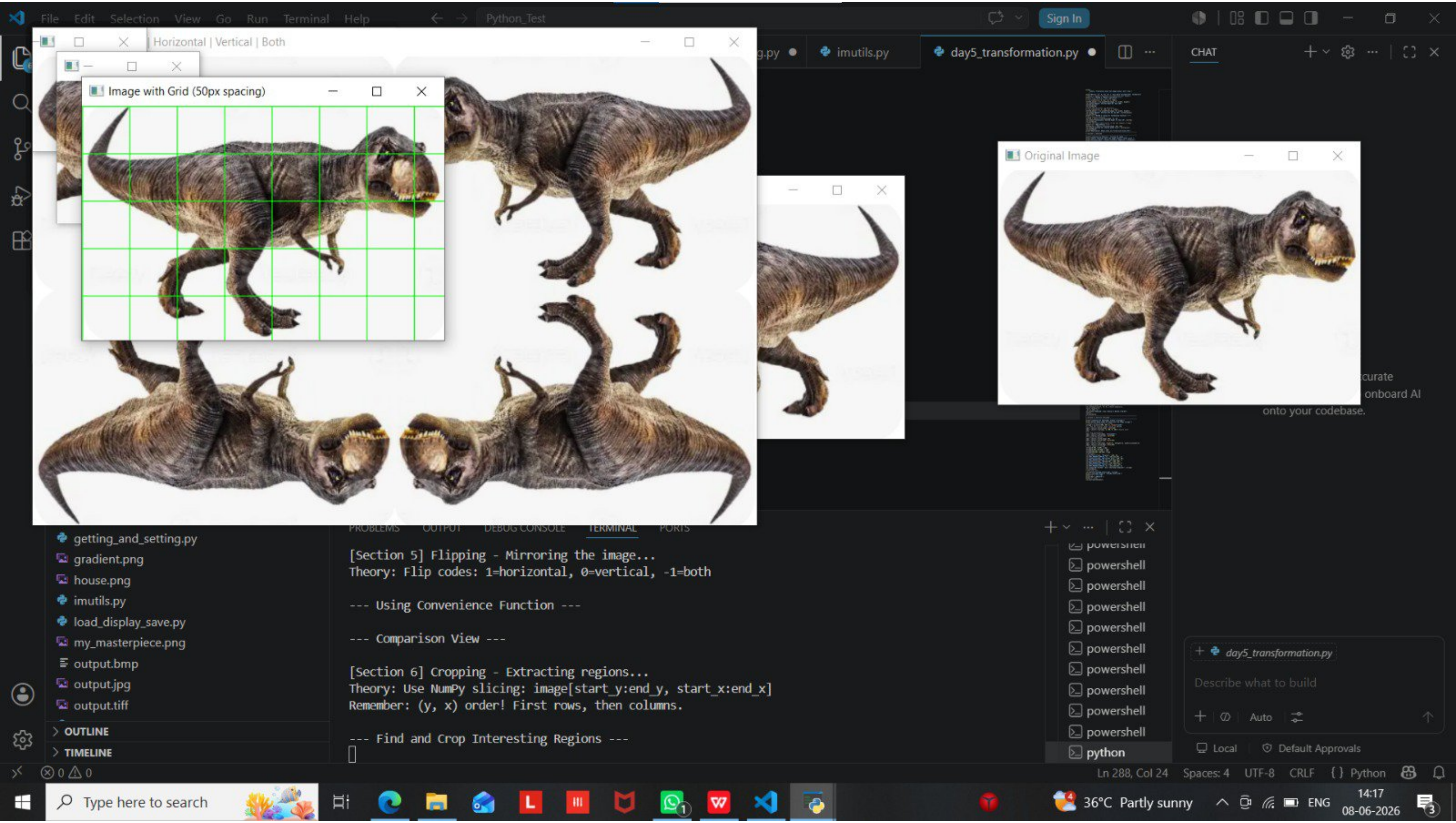
File Edit Selection View Go Run Terminal Help
Python_Test
Sign In
Original with Bounding Box
exercise2_panorama_prep.py
exercise3_rotation_animation.py
exercise4_smart_crop.py
CHAT
_smart_crop.py
exercise_4_smart_crop.py
port cv2
port numpy as np
port imutils
age = cv2.imread("test_pattern.png") # Use pattern image
Convert to grayscale and find edges
ay = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)
ges = cv2.Canny(gray, 50, 150)
Find contours
ntours, _ = cv2.findContours(edges, cv2.RETR_EXTERNAL,cv2.CHAIN_APPROX_SIMPLE)
contours:
Find largest contour by area
largest = max(contours, key=cv2.contourArea)
# Get bounding box
x, y, w, h = cv2.boundingRect(largest)
# Crop to bounding box with padding
padding = 20
y_start = max(0, y - padding)
y_end = min(image.shape[0], y + h + padding)
x_start = max(0, x - padding)
x_end = min(image.shape[1], x + w + padding)
exercise_1.py
exercise_2_target.py
exercise_2.py
exercise_3_chessboard.py
exercise_3.py
exercise_4_annotation.py
exercise_5_clock.py
exercise1_thumbnail_gallery.py
exercise2_panorama_prep.py
exercise3_rotation_animation.py
exercise4_smart_crop.py
first_image.py
getting_and_setting.py
gradient.png
house.png
imutils.cpython-314.pyc
imutils.py
load_display_save.py
OUTLINE
TIMELINE
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\jebron jason\Desktop\Python_Test> python exercise4_smart_crop.py
powershell
python
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
exercise4_smart_crop.py
Describe what to build
Auto
Local
Default Approvals
Ln 21, Col 47 Spaces: 4 UTF-8 CRLF { } Python
Type here to search
37°C Partly sunny
ENG
15:08
08-06-2026

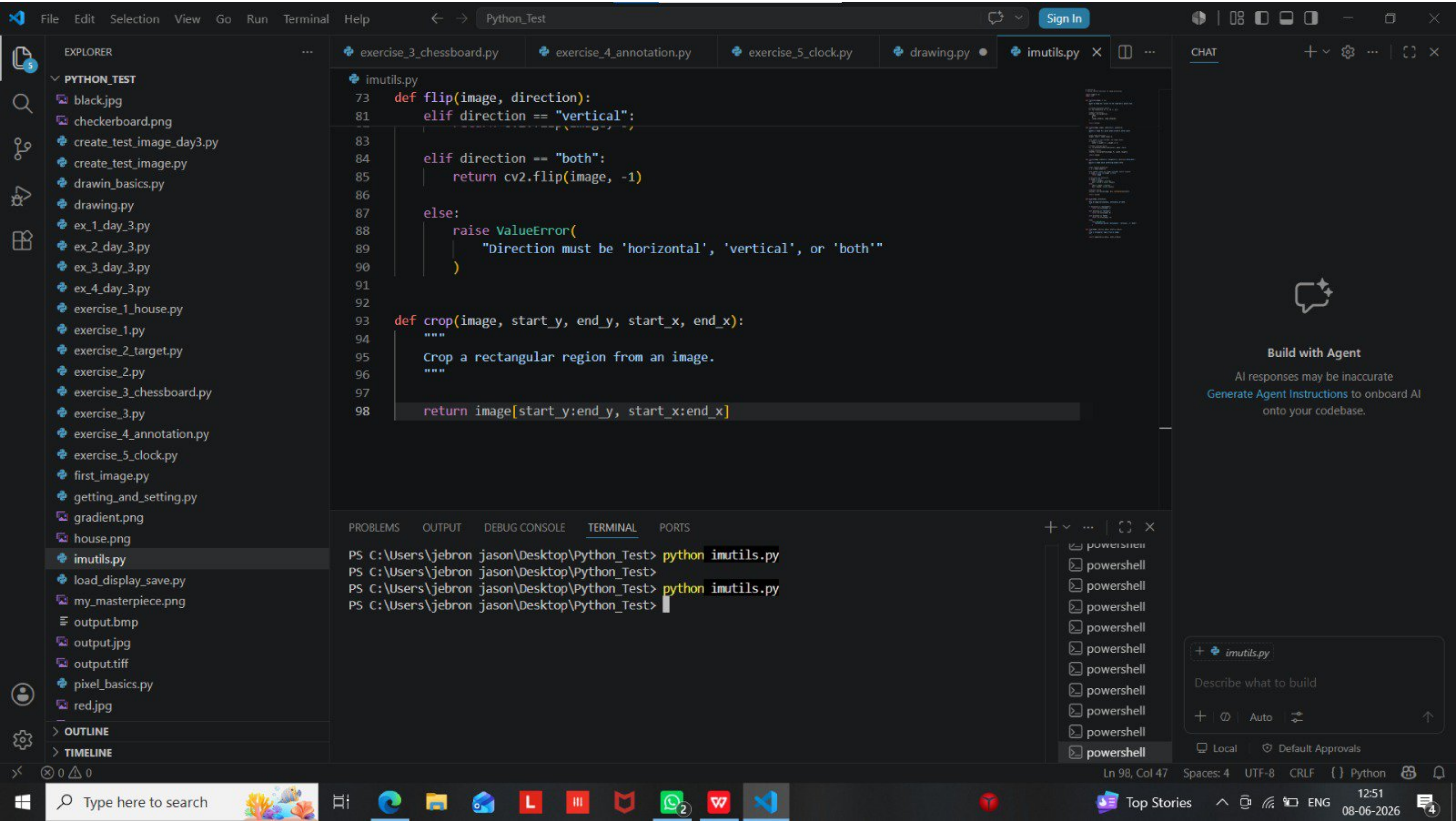

File Edit Selection View Go Run Terminal Help
Python_Test
Sign In
EXPLORER
PYTHON_TEST
__pycache__
black.jpg
create_test_image_day3.py
day5_transformation.py
drawing.py
ex_1_day_3.py
ex_2_day_3.py
ex_3_day_3.py
ex_4_day_3.py
exercise_1_house.py
exercise_1.py
exercise_2_target.py
exercise_2.py
exercise_3_chessboard.py
exercise_3.py
exercise_4_annotation.py
exercise_5_clock.py
exercise1_thumbnail_gallery.py
exercise2_panorama_prep.py
exercise3_rotation_animation.py
first_image.py
getting_and_setting.py
gradient.png
house.png
imutils.cpython-314.pyc
imutils.py
load_display_save.py
OUTLINE
TIMELINE
exercise1_thumbnail_gallery.py
exercise2_panorama_prep.py
exercise3_rotation_animation.py
Rotating Animation (Press any key to stop)
import cv2
import imutils
image = cv2.imread("test_image.png")
# Create a sequence of rotated images
for angle in range(0, 360, 15):
rotated = imutils.rotate(image, angle)
# Add angle text
cv2.putText(rotated, f"Angle: {angle} deg", (10, 30),
cv2.FONT_HERSHEY_SIMPLEX, 0.7, (0, 255, 0),2)
cv2.imshow("Rotating Animation (Press any key to stop)",rotated)
# Wait 100ms between frames
if cv2.waitKey(100) != -1:
break
cv2.destroyAllWindows()
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\jebron jason\Desktop\Python_Test> python exercise3_rotation_animation.py
File "C:\Users\jebron jason\Desktop\Python_Test\exercise3_rotation_animation.py", line 19
break
^^^^^
IndentationError: expected an indented block after 'if' statement on line 18
PS C:\Users\jebron jason\Desktop\Python_Test> python exercise3_rotation_animation.py
File "C:\Users\jebron jason\Desktop\Python_Test\exercise3_rotation_animation.py", line 19
break
^^^^^
IndentationError: expected an indented block after 'if' statement on line 18
PS C:\Users\jebron jason\Desktop\Python_Test> python exercise3_rotation_animation.py
PS C:\Users\jebron jason\Desktop\Python_Test> python exercise3_rotation_animation.py
CHAT
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
Describe what to build
Auto
Local
Default Approvals
Ln 18, Col 5 Spaces: 4 UTF-8 CRLF {} Python
Type here to search
37°C Partly sunny
ENG
15:03
08-06-2026



File Edit Selection View Go Run Terminal Help
Python_Test
Sign In
EXPLORER
PYTHON_TEST
__pycache__
black.jpg
checkerboard.png
create_test_image_day3.py
create_test_image.py
day5_transformation.py
drawin_basics.py
drawing.py
ex_1_day_3.py
ex_2_day_3.py
ex_3_day_3.py
ex_4_day_3.py
exercise_1_house.py
exercise_1.py
exercise_2_target.py
exercise_2.py
exercise_3_chessboa
exercise_3.py
exercise_4_annotatic
exercise_5_clock.py
exercise1_thumbnai
exercise2_panorama
first_image.py
getting_and_setting
gradient.png
house.png
imutils.cpython-314
imutils.py
load_display_save.py
my_masterpiece.png
OUTLINE
TIMELINE
imutils.py
day5_transformation.py
exercise1_thumbnail_gallery.py
exercise2_panorama_prep.py
CHAT
# exercise_2_panorama_prep.py
import numpy as np
import cv2
import imutils
# Create two shifted versions of an image to simulate panorama stitching
image = cv2.imread("test_image.png") # Use your image
# Create left and right shifted versions
left_shift = imutils.translate(image, -80, 0)
right_shift = imutils.translate(image, 80, 0)
# Create overlay to show overlap
overlay = image.copy()
overlay[0:image.shape[0], 0:80] =left_shift[0:image.shape[0], image.shape[1]-80:image.shape[1]
cv2.putText(overlay, "Overlap Region", (10, 50),
Panorama Preparation (Left | Original | Right)
Agent
be inaccurate
tions to onboard AI
odebase.
prep.py
PS C:\Users\jebron jason\Desktop\Python_Test> python exercise2_panorama_prep.py
powershell
python
Local
Default Approvals
Ln 12, Col 78 Spaces: 4 UTF-8 CRLF {} Python
Type here to search
37°C Partly sunny
ENG
14:55
08-06-2026


exercise1_thumbnail_gallery.py
for i, (name, transform) in enumerate(transformations):
col = i % 4
thumb = cv2.resize(transform(image), (thumb_size, thumb_size))
y_start = row * thumb_size
y_end = (row + 1) * thumb_size
x_start = col * thumb_size
x_end = (col + 1) * thumb_size
gallery[y_start:y_end, x_start:x_end] = thumb
cv2.putText(gallery, name, (x_start + 5, y_start + 20),
cv2.FONT_HERSHEY_SIMPLEX, 0.4, (255, 255,255), 1)
cv2.imshow("Thumbnail Gallery", gallery)
cv2.waitKey(0)
cv2.imwrite("thumbnail_gallery.png", gallery)
cv2.destroyAllWindows()
cv2.imshow("Thumbnail Gallery", gallery)
cv2.waitKey(0)
cv2.imwrite("thumbnail_gallery.png", gallery)
cv2.destroyAllWindows()
Thumbnail Gallery
Crop Center
Zoom In
PS C:\Users\jebron jason\Desktop\Python_Test> python exercise1_thumbnail_gallery.py
usage: exercise1_thumbnail_gallery.py [-h] -i IMAGE
exercise1_thumbnail_gallery.py: error: the following arguments are required: -i/--image
PS C:\Users\jebron jason\Desktop\Python_Test> python exercise1_thumbnail_gallery.py --image trex.png


File Edit Selection View Go Run Terminal Help
Python_Test
Sign In
Horizontal | Vertical | Both
Image with Grid (50px spacing)
Original Image
imutils.py
day5_transformation.py
CHAT
getting_and_setting.py
gradient.png
house.png
imutils.py
load_display_save.py
my_masterpiece.png
output.bmp
output.jpg
output.tiff
OUTLINE
TIMELINE
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
[Section 5] Flipping - Mirroring the image...
Theory: Flip codes: 1=horizontal, 0=vertical, -1=both
--- Using Convenience Function ---
--- Comparison View ---
[Section 6] Cropping - Extracting regions...
Theory: Use NumPy slicing: image[start_y:end_y, start_x:end_x]
Remember: (y, x) order! First rows, then columns.
--- Find and Crop Interesting Regions ---
powershell
python
onto your codebase.
Describe what to build
Auto
Local
Default Approvals
Ln 288, Col 24 Spaces: 4 UTF-8 CRLF { } Python
Type here to search
36°C Partly sunny
ENG
14:17
08-06-2026


File Edit Selection View Go Run Terminal Help
Python_Test
Sign In
EXPLORER
PYTHON_TEST
black.jpg
checkerboard.png
create_test_image_day3.py
create_test_image.py
drawin_basics.py
drawing.py
ex_1_day_3.py
ex_2_day_3.py
ex_3_day_3.py
ex_4_day_3.py
exercise_1_house.py
exercise_1.py
exercise_2_target.py
exercise_2.py
exercise_3_chessboard.py
exercise_3.py
exercise_4_annotation.py
exercise_5_clock.py
first_image.py
getting_and_setting.py
gradient.png
house.png
imutils.py
load_display_save.py
my_masterpiece.png
output.bmp
output.jpg
output.tiff
pixel_basics.py
red.jpg
OUTLINE
TIMELINE
exercise_3_chessboard.py
exercise_4_annotation.py
exercise_5_clock.py
drawing.py
imutils.py
def flip(image, direction):
    elif direction == "vertical":
    elif direction == "both":
        return cv2.flip(image, -1)
    else:
        raise ValueError(
            "Direction must be 'horizontal', 'vertical', or 'both'"
)
def crop(image, start_y, end_y, start_x, end_x):
    """
    Crop a rectangular region from an image.
    """
    return image[start_y:end_y, start_x:end_x]
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\jebron jason\Desktop\Python_Test> python imutils.py
PS C:\Users\jebron jason\Desktop\Python_Test>
PS C:\Users\jebron jason\Desktop\Python_Test> python imutils.py
PS C:\Users\jebron jason\Desktop\Python_Test>
powershell
CHAT
Build with Agent
AI responses may be inaccurate
Generate Agent Instructions to onboard AI onto your codebase.
imutils.py
Describe what to build
Auto
Local
Default Approvals
Ln 98, Col 47 Spaces: 4 UTF-8 CRLF { } Python
Type here to search
Top Stories
ENG
12:51
08-06-2026